HDOC DAY-44

Title: String Processing: Individual Word Reversal and Name Initials Extraction

1. Problem Statement

Write a menu-driven C program to perform the following string operations using separate functions:

1. Reverse each word in a sentence without changing the word order.

Input: What is special?, *Output:* tahw si ?laiceps

2. Print the initials of a name.

Input: Alexander Grahambell, *Output:* A.G.

3. Exit.

2. Theoretical Background & Algorithm Logic

- **Word-by-Word Reversal:**

The function scans the sentence, isolates individual words bounded by spaces or punctuation marks, and reverses each token internally using an index-swapping approach while keeping the original sequence of words intact.

- **Initials Extraction:**

The function iterates through the name string, detects character sequences following whitespace boundaries or starting at index zero, and outputs their uppercase equivalents appended with periods (.).

3. Test Case Table

Test Case	Selected Option	Input Data	Expected Output	Actual Output	Status
1	Reverse Words	"What is special?"	"tahw si ?laicep s"	"tahw si ?laicep s"	Pass
2	Print Initials	"Alexander Grahambell"	"A.G."	"A.G."	Pass
3	Print Initials	"Hardik Suhag"	"H.S."	"H.S."	Pass
4	Exit	Choice 3	Program termina	Program termina	Pass

Test Case	Selected Option	Input Data	Expected Output	Actual Output	Status
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4. Source Code Listing

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void read_string(const char *prompt, char str[], int max_size) {
   printf("%s", prompt);
   scanf(" %[^\n]", str);
}

void reverse_each_word(const char input[], char output[]) {
   char temp[500];
   strcpy(temp, input);
   output[0] = '\0';

   int i = 0;
   while (temp[i] != '\0') {
      int start = i;
      while (temp[i] != '\0' && !isspace((unsigned char)temp[i])) {
         i++;
      }
      int end = i - 1;

      char rev_word[100];
      int k = 0;
      for (int j = end; j >= start; j--) {
         rev_word[k++] = temp[j];
      }
```

```
        rev_word[k] = '\0';
        strcat(output, rev_word);

        if (temp[i] != '\0') {
            char space_str[2] = {temp[i], '\0'};
            strcat(output, space_str);
            i++;
        }
    }
}

void print_initials(const char name[]) {
    int i = 0;
    int new_word = 1;

    printf("Initials: ");
    while (name[i] != '\0') {
        if (isspace((unsigned char)name[i])) {
            new_word = 1;
        } else if (new_word) {
            printf("%c.", (char)toupper((unsigned char)name[i]));
            new_word = 0;
        }
        i++;
    }
    printf("\n");
}

int main(void) {
    char sentence[500], result[500], name[200];
    int choice;

    do {
        printf("\n1. Reverse Words\n2. Print Initials\n3. Exit\nChoice: ");
        if (scanf("%d", &choice) != 1) break;

        switch (choice) {
            case 1:
                read_string("Enter sentence: ", sentence, sizeof(sentence));
                reverse_each_word(sentence, result);
```

```
            printf("Result: %s\n", result);
            break;
        case 2:
            read_string("Enter name: ", name, sizeof(name));
            print_initials(name);
            break;
        case 3: printf("Exiting...\n"); break;
        default: printf("Invalid choice!\n"); break;
      }
   } while (choice != 3);

   return 0;
}
```

